

# GoPhish lab report — Local phishing simulation (MailHog)

## Objective

Demonstrate a safe phishing simulation workflow using GoPhish and a local SMTP sink (MailHog). The goal was to send a single test phishing email to a controlled address, verify delivery/click/submission tracking, and capture evidence for reporting.

## Environment & tools

- **GoPhish** — phishing simulation platform (admin UI + phish server).
  - Admin UI: `https://127.0.0.1:3333` (default)
  - Phish server: `http://127.0.0.1` (or `http://127.0.0.1:<port>` if different)
- **MailHog (Docker)** — local SMTP catcher to capture outgoing email without sending externally.
  - SMTP: `127.0.0.1:1025`
  - Web UI: `http://127.0.0.1:8025`
- **Docker Desktop** — ran MailHog container:
- `docker run -d -p 1025:1025 -p 8025:8025 --name mailhog mailhog/mailhog`
- Browser: Chrome

### 1.Started MailHog (Docker) to capture SMTP:

```
docker run -d -p 1025:1025 -p 8025:8025 --name mailhog
mailhog/mailhog
```

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\SHARIKACOMPUTERS> docker run -d -p 1025:1025 -p 8025:8025 --name mailhog mailhog/mailhog
Unable to find image 'mailhog/mailhog:latest' locally
latest: Pulling from mailhog/mailhog
368ee3bcd1dbb: Pull complete
f17c8f1adafb: Pull complete
03954754c53a: Pull complete
df20fa9351a1: Pull complete
ed8968b2872e: Pull complete
60493946972a: Pull complete
a92cc7c5fd73: Pull complete
Digest: sha256:8d76a3d4ffa32a3661311944007a415332c4bb855657f4f6c57996405c009bea
Status: Downloaded newer image for mailhog/mailhog:latest
0d06f5120a08bd8f16b99f4d7e04b836779aca5ccdc410d075a3ba7b9aac671a
PS C:\Users\SHARIKACOMPUTERS> docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
0d06f5120a08        mailhog/mailhog    "MailHog"          3 minutes ago       Up 3 minutes       0.0.0.0:1025->1025/tcp, [::]:1025->1025/tcp, 0.0.0.0:8025->8025/tcp, [::]:8025->8025/tcp
PS C:\Users\SHARIKACOMPUTERS>
```

## 2.Started GoPhish (gophish.exe / ./gophish) and noted terminal output:

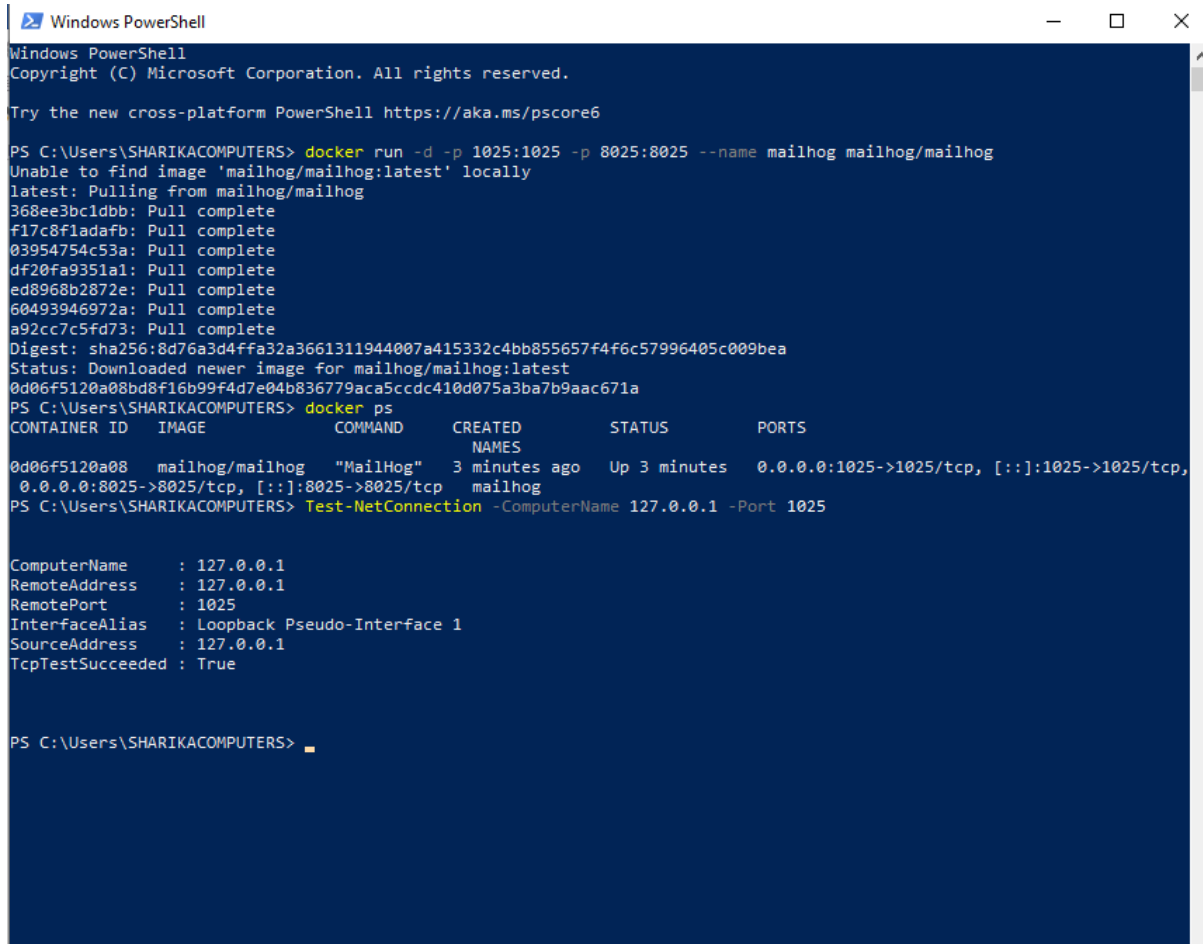
- Admin server (UI): <https://127.0.0.1:3333>
- Phish server listening (e.g. 0.0.0.0:80) → used <http://127.0.0.1> as campaign URL

```
C:\Users\SHARIK~1\AppData\Local\Temp\Rar$EXa9692.42751\gophish.exe
OK 20160225173824_0.1.2_capture_credentials.sql
OK 20160227180335_0.1.2_store-smtp-settings.sql
OK 20160317214457_0.2_redirect_url.sql
OK 20160605210903_0.2_campaign_scheduling.sql
OK 20170104220731_0.2_result_statuses.sql
OK 20170219122503_0.2.1_email_headers.sql
OK 20170827141312_0.4_utc_dates.sql
OK 20171027213457_0.4.1_maillogs.sql
OK 20171208201932_0.4.1_next_send_date.sql
OK 20180223101813_0.5.1_user_reporting.sql
OK 20180524203752_0.7.0_result_last_modified.sql
OK 20180527213648_0.7.0_store_email_request.sql
OK 20180830215615_0.7.0_send_by_date.sql
OK 20190105192341_0.8.0_rbac.sql
OK 20191104103306_0.9.0_create_webhooks.sql
OK 20200116000000_0.9.0_imap.sql
OK 20200619000000_0.11.0_password_policy.sql
OK 20200730000000_0.11.0_imap_ignore_cert_errors.sql
OK 20200914000000_0.11.0_last_login.sql
OK 20201201000000_0.11.0_account_locked.sql
OK 20220321133237_0.4.1_envelope_sender.sql
time="2025-10-28T17:04:56+05:30" level=info msg="Please login with the username admin and the password ddeae1ca42adc444"
time="2025-10-28T17:04:56+05:30" level=info msg="Starting IMAP monitor manager"
time="2025-10-28T17:04:56+05:30" level=info msg="Starting phishing server at http://0.0.0.0:80"
time="2025-10-28T17:04:56+05:30" level=info msg="Background Worker Started Successfully - Waiting for Campaigns"
time="2025-10-28T17:04:56+05:30" level=info msg="Creating new self-signed certificates for administration interface"
time="2025-10-28T17:04:56+05:30" level=info msg="Starting new IMAP monitor for user admin"
time="2025-10-28T17:04:56+05:30" level=info msg="TLS Certificate Generation complete"
time="2025-10-28T17:04:56+05:30" level=info msg="Starting admin server at https://127.0.0.1:3333"
```

### 3. Test that port 1025 is reachable from Windows:

```
Test-NetConnection -ComputerName 127.0.0.1 -Port 1025
```

You want `TcpTestSucceeded` : True. If False, MailHog is not running or the port mapping is different.



```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\SHARIKACOMPUTERS> docker run -d -p 1025:1025 -p 8025:8025 --name mailhog mailhog/mailhog
Unable to find image 'mailhog/mailhog:latest' locally
latest: Pulling from mailhog/mailhog
368ee3bc1dbb: Pull complete
f17c8f1adafb: Pull complete
03954754c53a: Pull complete
df20fa9351a1: Pull complete
ed8968b2872e: Pull complete
60493946972a: Pull complete
a92cc7c5fd73: Pull complete
Digest: sha256:8d76a3d4ffa32a3661311944007a415332c4bb855657f4f6c57996405c009bea
Status: Downloaded newer image for mailhog/mailhog:latest
0d06f5120a08bd8f16b99f4d7e04b836779aca5ccdc410d075a3ba7b9aac671a
PS C:\Users\SHARIKACOMPUTERS> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
0d06f5120a08   mailhog/mailhog "MailHog"               3 minutes ago Up 3 minutes  0.0.0.0:1025->1025/tcp, [::]:1025->1025/tcp, 0.0.0.0:8025->8025/tcp, [::]:8025->8025/tcp mailhog
PS C:\Users\SHARIKACOMPUTERS> Test-NetConnection -ComputerName 127.0.0.1 -Port 1025

ComputerName      : 127.0.0.1
RemoteAddress     : 127.0.0.1
RemotePort        : 1025
InterfaceAlias    : Loopback Pseudo-Interface 1
SourceAddress     : 127.0.0.1
TcpTestSucceeded  : True

PS C:\Users\SHARIKACOMPUTERS>
```

### 4. Configured GoPhish Sending Profile:

- Host: 127.0.0.1:1025 (MailHog SMTP)
- Username / Password: **left blank**
- From: Test Sender <you@local.test>
- Tested connection → **Success**

← → ↻ Not secure https://127.0.0.1:3333/sending\_profiles

admin Guest

**gophish**

Dashboard  
Campaigns  
Users & Groups  
Email Templates  
Landing Pages  
**Sending Profiles**  
Account Settings  
User Management Admin  
Webhooks Admin  
User Guide  
API Documentation

**New Sending Profile**

Name:  
MailHog-local

Interface Type:  
SMTP

SMTP From: ⓘ  
Test Sender <you@local.test>

Host:  
127.0.0.1:1025

Username:  
Username

Password:  
Password

## 5. Created Group with you@local . test

← → ↻ Not secure https://127.0.0.1:3333/groups

**gophish**

Dashboard  
Campaigns  
**Users & Groups**  
Email Templates  
Landing Pages  
Sending Profiles  
Account Settings  
User Management Admin  
Webhooks Admin  
User Guide  
API Documentation

**New Group**

Name:  
Test Group

[+ Bulk Import Users](#) [Download CSV Template](#)

Test User you@local.test optional [+ Add](#)

Show 10 entries Search:

| First Name                 | Last Name | Email | Position |
|----------------------------|-----------|-------|----------|
| No data available in table |           |       |          |

Showing 0 to 0 of 0 entries

Previous Next

Close [Save changes](#)

## 6. Created Email Template and Landing Page (HTML pasted into GoPhish UI)

The first screenshot shows the 'Edit Template' form in the GoPhish UI. The browser address bar indicates the URL is `https://127.0.0.1:3333/templates`. The form includes fields for Name (Test Template), Envelope Sender (Test Sender <you@local.test>), and Subject (Security Update — Action Required). The HTML tab is selected, showing a placeholder for a first name and a link for account review. The left sidebar shows the navigation menu with 'Email Templates' highlighted.

The second screenshot shows the 'New Landing Page' form in the GoPhish UI. The browser address bar indicates the URL is `https://127.0.0.1:3333/landing_pages`. The form includes a Name field (Test Login) and an 'Import Site' button. The HTML tab is selected, showing a form for account verification with fields for email and password. The left sidebar shows the navigation menu with 'Landing Pages' highlighted.

**Edit Template**

Name: Test Template

Import Email

Envelope Sender: Test Sender <you@local.test>

Subject: Security Update — Action Required

Text HTML

<p>Hi {{.FirstName}},</p>

<p>We detected unusual activity on your account. Please <a href="{{.URL}}">click here to review</a>.</p>

**New Landing Page**

Name: Test Login

Import Site

HTML

<!doctype html>

<html>

<body>

<h3>Account Verification</h3>

<form method="POST">

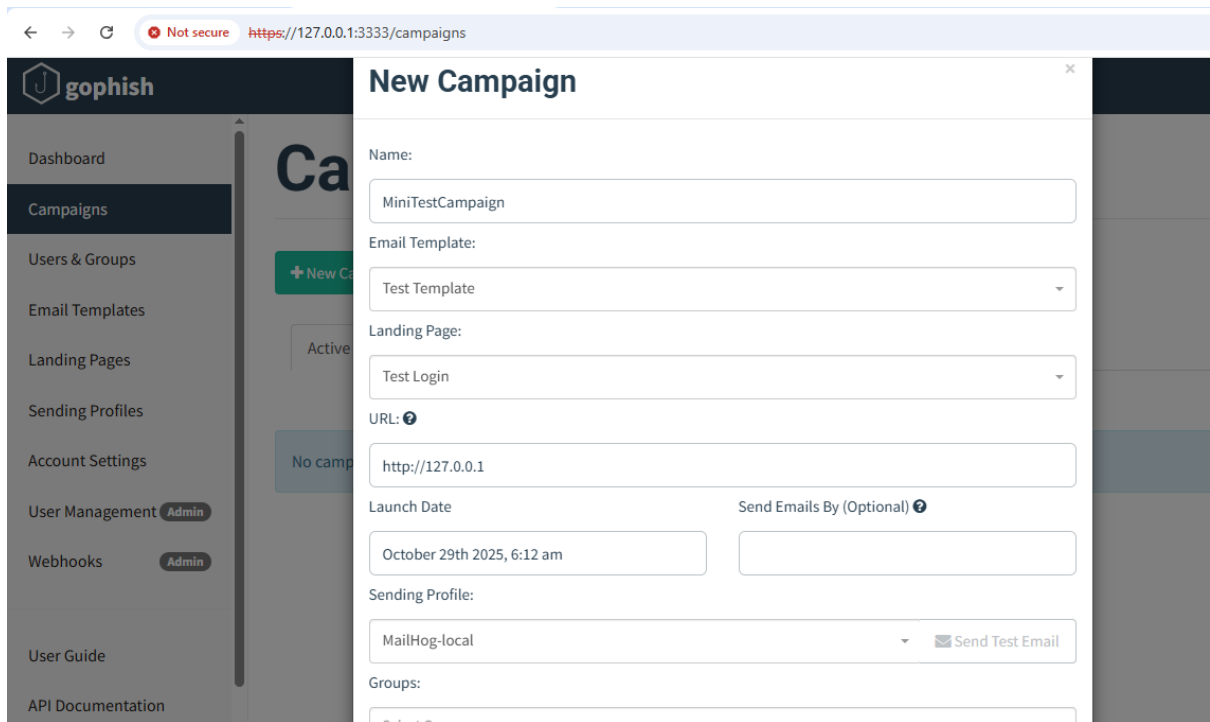
<label>Email</label><br>

<input name="email" /><br><br>

<label>Password</label><br>

Capture Submitted Data

## 7. Launched campaign (Mini Test Campaign)

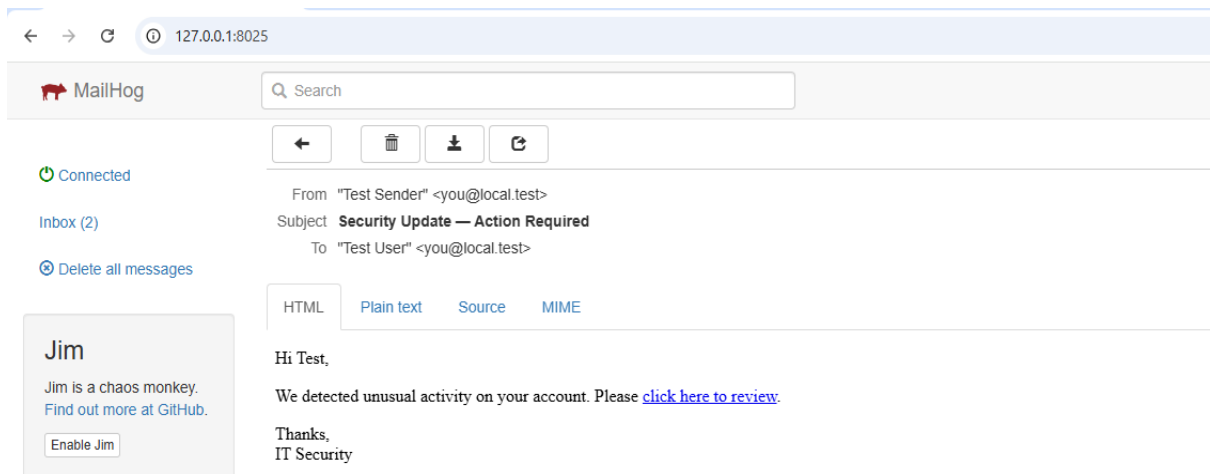


The screenshot shows the 'New Campaign' form in the Gophish web interface. The browser address bar shows 'https://127.0.0.1:3333/campaigns'. The form includes the following fields:

- Name:** MiniTestCampaign
- Email Template:** Test Template
- Landing Page:** Test Login
- URL:** http://127.0.0.1
- Launch Date:** October 29th 2025, 6:12 am
- Send Emails By (Optional):** (empty)
- Sending Profile:** MailHog-local
- Groups:** (empty)

There is a 'Send Test Email' button next to the Sending Profile dropdown. The left sidebar shows the Gophish navigation menu with options like Dashboard, Campaigns, Users & Groups, Email Templates, Landing Pages, Sending Profiles, Account Settings, User Management, Webhooks, User Guide, and API Documentation.

## 8. Verified delivery in MailHog UI: http://127.0.0.1:8025



The screenshot shows the MailHog web interface at 'http://127.0.0.1:8025'. The interface displays a received email with the following details:

- From:** "Test Sender" <you@local.test>
- Subject:** Security Update — Action Required
- To:** "Test User" <you@local.test>

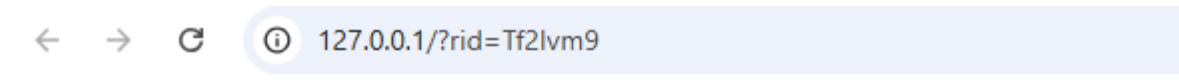
The email content is displayed in 'Plain text' format:

Hi Test,

We detected unusual activity on your account. Please [click here to review](#).

Thanks,  
IT Security

On the left sidebar, there is a 'Connected' status, an 'Inbox (2)' link, a 'Delete all messages' button, and a 'Jim' notification card that says 'Jim is a chaos monkey. Find out more at GitHub.' with an 'Enable Jim' button.

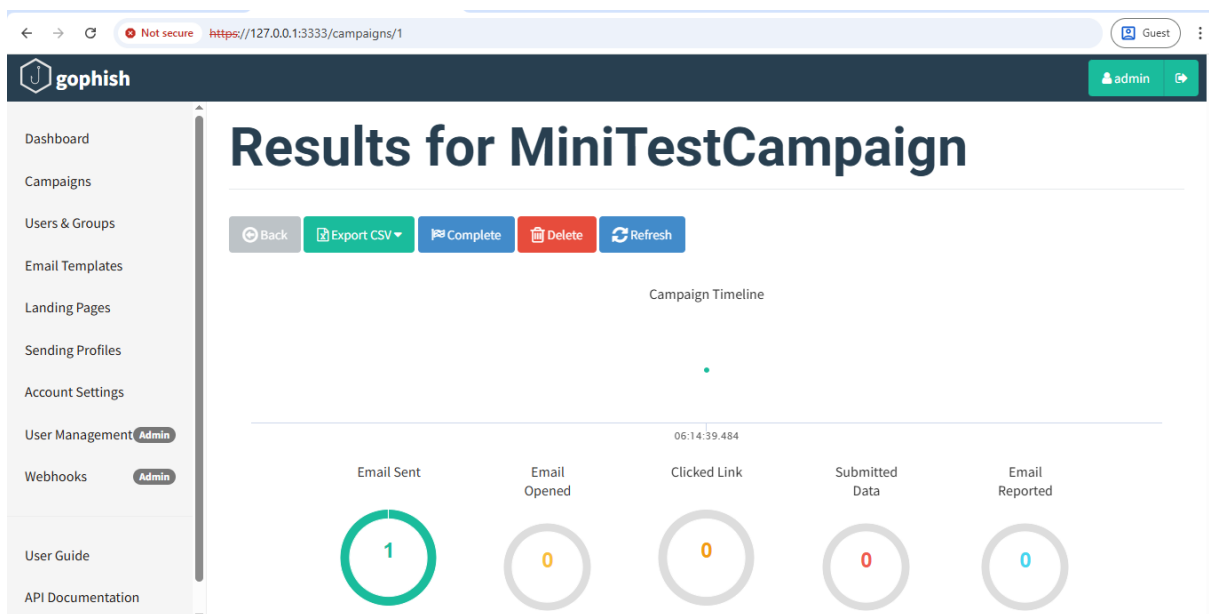


## Account Verification

Email

Password

Submit



## Analysis & notes

- Using MailHog prevented any real external email delivery — this ensured safe, contained testing.
- Tracking succeeded (delivery → click → submission), demonstrating the full GoPhish workflow in a controlled environment.
- Because MailHog captures any sender address, no SPF/DKIM/DMARC or relay restrictions applied; using a real SMTP provider would require proper credentials and could result in deliverability differences.

## Conclusion

The simulated phishing campaign was successfully executed locally. GoPhish captured click and form submission events as expected.